Toward a four-dimensional Yang–Mills construction

Notebook 37: the actual logarithm and finite local ultraviolet tuning

Research proof draft

17 October 2026

Abstract

We complete the leading-determinant comparison left open in Notebook 36, under the explicitly stated classical structural and local quantum inputs of this working series. The actual coefficient satisfies $\beta(\mathcal{D}_{M,K}) = 22\alpha_g \log K + O(1)$, uniformly in preceding depth and in all applicable joint infinite structural systems. Here $\alpha_g = C_H/(48\pi^2)$ uses the original color normalization.

The proof has three new steps. First, source-localized sectorial resolvents remove the soft infrared penalties through physical time one with an integrable $O(u)$ error. Second, nonlinear Ward identities through field degree four compare the actual effective-action heat coefficient with the Wilson coefficient with error $O(s^{-1})$. The left-relative chart-current terms are retained. Third, a direct Wilson-to-continuum comparison gives an integrable $O(s^{-1/2})$ error and fixes the coefficient without importing a universality assertion for an unmatched regulator.

The positive margin now supplies the previously missing hypothesis of the finite local shooting theorem. It yields arbitrarily long finite weak local histories with a fixed weak endpoint and bare coupling of order the inverse square root of their length. This is a theorem about the homogeneous local recursion. Full Wilson measure matching across all field regions, sourced continuum limits, and a nontrivial reconstructed Yang–Mills theory remain unproved.

Scope and the proof dependency

This is a research proof draft for mathematical review. Its new arguments are given below; the previously constructed classical actions, exact determinant identities, localized vertices and finite local recursion are inputs from the earlier notebooks. Internal mathematical audits and successful typesetting do not certify those upstream proofs. The main result is not a completed four-dimensional existence theorem. The deposited construction principally concerns $SU(2)$. General color-factor formulas below do not by themselves establish all upstream construction inputs for every compact simple group.

Current coordinates use counting pairings. The primitive ratio $\ell > 1$ is fixed, $K = \ell^k$, and final physical spacing is K^{-1}. The preceding classical depth is M. The notation $[t^2]$ is a Taylor coefficient, whereas the Maxwell functional is $\beta = 2[t^2]$ on the normalized affine probe. For a Cartan element with $\tau(H^2) = 1$, put

$$C_H = \sum_{q \in \mathrm{spec}(\mathrm{ad}_H)} q^2, \qquad \alpha_g = C_H/(48\pi^2).$$

Both signs of a charge are included in this adjoint trace; its complexification introduces no extra multiplicity.

The infinite spatial estimates refer to joint structural systems containing all required old-depth shifts and endpoint ratios. Their constants are independent of the chosen system. Exact finite-period determinant algebra and finite flat spectral bounds are distinguished from continuous-momentum identities of these infinite kernels. The finite local conclusion is recovered by the separate compactness and moment-transfer theorem; no finite winding identity is inferred from an infinite Fourier expansion.

The proof chain is

$$\begin{gathered}\text{Notebook 36: exact soft reduction and bounded heat ends}\\ \Downarrow\\ \text{soft penalty removal} \longrightarrow \text{actual base versus Wilson}\\ \longrightarrow \text{Wilson versus continuum} \longrightarrow \text{positive actual logarithm}\\ \longrightarrow \text{finite local transfer and shooting.}\end{gathered}$$

The final section records the interfaces from this local theorem to the original existence objective.

1 Removing the soft penalties up to physical time one

Work in dimension four, in current counting coordinates, on the chosen infinite structural system. Flat spatial bounds also pass to explicit periodizations by summing the exponentially decaying lifted kernels. This is not a linear-weight similarity on a finite torus. The affine coefficient itself is taken on the infinite structural system, not as a literal periodic magnetic flux; arbitrary actual finite winding kernels are not identified with those periodizations. Set $\mu = K^{-1}$. The affine background below means only the first two formal polynomial jets of

$$U_\alpha(t;x) = \exp(itx_\nu H/K^2), \qquad \alpha \neq \nu.$$

No nonzero unbounded field is inserted in a common analytic ball.

1.1 Actual families and the required inputs

Use the actual effective classical Hessians and moving-reference maps:

$$\begin{aligned} B_s(A) &= D_A^T D_A, & B_v(A) &= S''_M(e^{i\mu A}) + D_A D_A^T,\\ T_s(A) &= B_s(A) + \mu^2 J_A J_A^T, & \widetilde{V}(A) &= B_v(A) + \bar{C}_A^T F_A \bar{C}_A,\\ F_A &= aI - S''_{MK}(M_K(e^{i\mu A})). \end{aligned} \tag{1}$$

All Hessians are left-relative, including their coordinate-current terms. The auxiliary vector $\widetilde{V}$ has the already established first two affine jets of the actual soft vector; equality off shell is not asserted.

For either pair $(W,B) = (T_s, B_s)$ or $(W,B) = (\widetilde{V}, B_v)$, put $X_E = K^2 E$ and write

$$X_W = X_B + C_A^T F_A C_A, \qquad (C_A, F_A) = \begin{cases} (J_A^T, I), & \text{scalar},\\ (K\bar{C}_A,\ aI - S''_{MK}(M_K(e^{i\mu A}))), & \text{vector}. \end{cases} \tag{2}$$

The output fiber of C_A has a fixed color/site or color/bond dimension per final block. Choose the fixed $a \geq R_q + 1$, so that $F_0 \geq I$ in the vector case.

The proof uses the following established inputs, with constants uniform in M, K:

1. The actual flat q_M has an exponential current-coordinate kernel, analytic double zero, Maxwell lower bound and uniform second moments. Hence, with the componentwise physical gradient $\nabla = Kd_0$ and $H_0 = I + \nabla^T \nabla$,
$$cH_0 \leq I + X_{B,0} \leq CH_0. \tag{3}$$

2. The repaired covariant-Hessian theorem gives first-two physical C^1 vertices, with complete smooth source collars and full rooted operator bounds. The scaled gauge-generator products have the same property.

3. The physical maps $K\bar{C}_A$, J_A and their first-two background derivatives have uniform whole-block operator bounds and fixed physical range. The actual coarse Hessian has uniform rooted source kernels. Consequently $C_A^T F_A C_A$ has uniformly bounded zero-order physical vertices and full rooted source bounds.

These inputs concern actual effective actions and corrected tangents. No bare Wilson replacement, full uniform nonlinear physical domain, or arbitrary high-frequency bounded-source estimate is used.

Theorem 1.1. *Under these inputs, if P_0 is the appropriate fine scalar or bond projector onto final block zero, then*

$$\left|[t^2]\,\mathrm{Tr}\,P_0\big(e^{-uX_W(t)}-e^{-uX_B(t)}\big)\right|\le Cu,\qquad 0<u\le 1, \tag{4}$$

uniformly in old depth M and endpoint K. Equivalently,

$$\left|[t^2]\,\mathrm{Tr}\,P_0\big(e^{-sW(t)}-e^{-sB(t)}\big)\right|\le C\frac{s}{K^2},\qquad 0<s\le K^2. \tag{5}$$

In particular the absolute ds/s integral of the difference on this entire band is bounded uniformly.

1.2 Shifted flat resolvents with physical spatial weights

At zero the penalty in (2) is nonnegative and bounded uniformly. Thus (3) holds for W too. Fix $E = B$ or W and set

$$\mathcal{B}_0 = I + X_{E,0},\qquad R_z = (\mathcal{B}_0 - z)^{-1}.$$

Use a sectorial contour with rays $\arg z = \pm\pi/4$, $|z|\ge 1/4$, joined by the circular arc of radius $1/4$ to the left of the positive spectrum. The spectral theorem and energy comparison give

$$\begin{aligned}\|R_z\| &\le C(1+|z|)^{-1},\\ \|\nabla R_z\| + \|R_z\nabla^T\| &\le C(1+|z|)^{-1/2},\\ \|\nabla R_z\nabla^T\| &\le C.\end{aligned} \tag{6}$$

For example, $\|(\mathcal{B}_0+|z|)^{1/2}R_z(\mathcal{B}_0+|z|)^{1/2}\|\le C$ on this contour.

Only linear physical weights are needed for the nonlocal flat kernel. Conjugate by $e^{\theta\cdot x/K}$, with sufficiently small real θ. The analytic double zero gives the fine-symbol estimate

$$\|q_M(p+i\mu\theta)-q_M(p)\|\le C\{\mu|\theta|\sqrt{\delta(p)}+\mu^2|\theta|^2\}.$$

After multiplication by K^2, this is a small form perturbation relative to H_0. The local gradient term has the same estimate. The coarse correction also does: its physical legs have bounded norm and fixed range, and its middle flat kernel has a uniform coarse analytic strip. The estimate remains true relative to $H_0+|z|$. A weighted resolvent Neumann series therefore proves (6) uniformly after conjugation. Commuting a weight through ∇ adds only $O(|\theta|)$ and is covered by the scalar resolvent estimate.

For each pair of blocks choose the signs of θ along their separation. The restricted versions of all three bounds then acquire $e^{-md(i,j)}$, for a common $m>0$. Internal block diameters change only the constant. These are whole-block Hilbert operator estimates, with no fine-entry trace or growing-fiber count. At fixed K the operators are bounded lattice operators; finite-support approximation justifies the weighted kernels. All constants are uniform in K.

1.3 At most one unbounded derivative per vertex

Let $X_j = [t^j]X_E(tA)$, $j = 1, 2$, for a complete physical C^1 direction of norm at most a_*. The repaired Hessian decomposition, multiplied by K^2, gives

$$X_j = \nabla^T \overline{Z}_j + \overline{Z}_j^T \nabla + \overline{E}_j, \qquad \|\overline{Z}_j\| + \|\overline{E}_j\| \le C_j a_*^j. \tag{7}$$

Indeed $\overline{Z}_j = KZ_j$ and $\overline{E}_j = K^2 E_j$; the improved $\overline{Z}_2 = O(K^{-1})$ is not needed. The scaled gauge-generator terms have this form, and the physical coarse penalty contributes bounded zero-order terms. Mixed second coefficients follow by polarization.

For directions in smooth physical collars x, y, each factor retains an exponential connecting-tree bound including its operator endpoints and every source. The significant fact is that every summand in (7) has at most one unbounded physical derivative. An arbitrary second-order form bound would not give the following integrable resolvent powers.

Only two jets are needed. If a full uniform nonlinear domain has not been provided, use the polynomial family $X(t) = X_0 + tX_1 + t^2 X_2$, or its mixed quadratic version. Equation (7) gives a small common parameter disk on which $I + X(t)$ is sectorial with the chosen contour. At each fixed K this is ordinary holomorphic functional calculus for bounded operators. Its first-two semigroup jets equal the actual ones because only X_0, X_1, X_2 enter their resolvent derivatives. Thus this device adds no nonlinear-domain hypothesis.

1.4 Rooted semigroup derivatives

Differentiated resolvents satisfy

$$\begin{aligned} \delta_x R_z &= -R_z X_x R_z, \\ \delta_x \delta_y R_z &= R_z X_x R_z X_y R_z + R_z X_y R_z X_x R_z - R_z X_{xy} R_z. \end{aligned} \tag{8}$$

One first-order insertion in (7) costs $(1 + |z|)^{-3/2}$ by (6); its zero-order part costs $(1 + |z|)^{-2}$. Two first-order insertions cost at worst $(1 + |z|)^{-2}$. If both derivatives face the middle resolvent, that piece costs $O(1)$ and both exterior resolvents cost $(1 + |z|)^{-1}$. In the other arrangements there are two half-power energy legs and one full resolvent. The single X_{xy} insertion again has exponent $-3/2$.

Insert block projections between the already defined factors $R_z \nabla^T$, ∇R_z and the bounded vertex factors. No sharp block cutoff is commuted through a derivative. The weighted resolvent and vertex bounds then give, after an exponent reserve,

$$\begin{aligned} \|\delta_x (R_z)_{ij}\| &\le C(1 + |z|)^{-3/2} e^{-m\tau(i,j,x)}, \\ \|\delta_x \delta_y (R_z)_{ij}\| &\le C\{(1 + |z|)^{-3/2} + (1 + |z|)^{-2}\} e^{-m\tau(i,j,x,y)}. \end{aligned} \tag{9}$$

Every intermediate sum is over block operator norms; a smaller exponent pays the finite convolution multiplicities.

Integrate these formulas on the Dunford contour for $e^{-u\mathcal{B}_0}$. The rays have positive real part, the joining arc has fixed length, and both powers in (9) are integrable at infinity, uniformly for $0 < u \le 1$. Since $e^{-uX} = e^u e^{-u(I+X)}$, one obtains

$$\begin{aligned} \|\delta_x (e^{-uX})_{ij}\| &\le C e^{-m'\tau(i,j,x)}, \\ \|\delta_x \delta_y (e^{-uX})_{ij}\| &\le C e^{-m'\tau(i,j,x,y)}. \end{aligned} \tag{10}$$

For instance the ray integrals of $(1 + \lambda)^{-3/2}$ and $(1 + \lambda)^{-2}$ are bounded by fixed constants even when the factor $e^{-u\lambda/\sqrt{2}}$ is discarded.

The undifferentiated semigroup has a uniform exponential restricted block bound too. The small linear-weight form perturbation retains the positive gradient energy, leaving numerical range bounded below by $-C$. Its real-time norm is at most e^{Cu}, and the weight gives the block decay. This uses only the flat generator.

This resolvent argument avoids the nonintegrable middle-time factor that an overly crude split of a second Duhamel insertion would produce. It also retains the source roots which a global Cauchy bound alone would lose.

1.5 Affine reconstruction and the coarse trace

Use a fixed smooth physical partition for the affine potential. Its direction in collar x has physical C^1 norm at most $C(1+|x|)$. Summing its first and mixed second directions against (10) gives, for $n = 0, 1, 2$,

$$\|([t^n]e^{-uX(t)})_{ij}\| \le C(1+|i|+|j|)^p e^{-m''d(i,j)}, \qquad 0 < u \le 1. \tag{11}$$

At $u = 0$ use the identity and zero differentiated coefficients. The sums are absolutely convergent polynomial-jet sums, not evaluations at a nonzero unbounded field.

The physical C_A, F_A in (2) have the same polynomial block bounds; C_A has fixed physical range and order-one norm. In the vector case ordinary averaging sends the affine connection to the coarse affine connection, so the actual coarse Hessian supplies the required polynomial/exponential F kernels.

Retain P_0 in the exact physical-time Duhamel trace identity:

$$\operatorname{Tr}_f P_0(e^{-uX_W} - e^{-uX_B}) = -\int_0^u \operatorname{Tr}_c[FCe^{-vX_B}P_0e^{-(u-v)X_W}C^T]\,dv. \tag{12}$$

The trace on the right sums all coarse diagonal blocks. Its degree-two coefficient has finitely many ordered terms. Equation (11) and the C, F estimates connect both coarse endpoints to the retained root block. Their absolute sum is bounded by a fixed multiple of

$$\sum_{Y,Z}(1+|Y|+|Z|)^p e^{-m'''(|Y|+|Z|+d(Y,Z))} < \infty.$$

Each diagonal trace is in a fixed-dimensional coarse output fiber. A product through the fine block-zero Hilbert space costs that coarse rank, not its growing fine dimension. Thus the integrand coefficient is bounded uniformly in u, v, M, K, including the endpoints $v = 0, u$. The integral has length u, proving (4).

For compact source truncations the cyclic trace step is ordinary finite-rank cyclicity with P_0 retained. The displayed absolute rooted sum gives the infinite limit and permits coefficient extraction. This completes the proof of Theorem 1.1.

1.6 Consequence and remaining comparison

Equation (5) implies

$$\int_0^{K^2} \left|[t^2]\operatorname{Tr} P_0(e^{-sW(t)} - e^{-sB(t)})\right| \frac{ds}{s} \le C. \tag{13}$$

Together with the coarse determinant reductions, this removes the soft penalties from the fine-to-physical-unit cutoff heat coefficient up to a uniform finite part. The vector conclusion is actual through the established affine matching; the scalar family is actual directly.

The remaining middle-band comparison concerns the translation-invariant actual base $S''_M + DD^T$ and the ordinary scalar Laplacian. This section does not replace that effective Hessian by a fixed bare reference, compute its logarithmic coefficient, or prove the separate physical-time tail beyond one. A theorem for a fixed bare integrand is not imported as a uniform statement in the present family.

2 The complete mean-zero scalar coefficient

The exact factorization of Notebook 36 reads

$$\det \widehat{G} = \det T_s \det M_s, \qquad T_s = D_A^T D_A + K^{-2} J_A J_A^T.$$

Here $M_s = K^{-2} J_A^T T_s^{-1} J_A$ has one fixed color fiber per final block. Its actual Maxwell coefficient is uniformly bounded. The normalization is $\beta(F) = 2[t^2]F$ on the specified one-color affine probe; normalization at the flat field contributes zero.

Theorem 2.1 (Mean-zero scalar logarithm)**.** *Under the actual structural and affine-identification hypotheses of Notebooks 35–36 and Theorem 1.1,*

$$\left|\beta(\mathcal{N}_0 \log\det \widehat{G}) - 2\alpha_g \log K\right| \le C, \qquad \alpha_g = \frac{C_H}{48\pi^2}. \tag{14}$$

The bound is uniform in preceding depth and in every applicable joint infinite structural system.

Proof. The scalar part of the infrared-tail theorem in Notebook 36 bounds the integral over $s > K^2$. Theorem 1.1 removes the scalar soft penalty on $0 < s \le K^2$ with a bounded integrated error. The coarse factor M_s contributes a further bounded error. The exact unpinned scalar heat calculation of Notebook 34 therefore gives

$$\beta(\mathcal{N}_0 \log\det \widehat{G}) = \int_0^{K^2} h_g(s)\,\frac{ds}{s} + O(1), \qquad h_g(s) = \frac{C_H}{3} s^2 e^{-8s} I_0(2s)^2 I_1(2s)^2. \tag{15}$$

The definition of h_g includes the factor -2 multiplying the scalar heat coefficient. The minus sign comes from the heat representation of the logarithm, and the factor two from β. No additional real/complex color multiplicity is introduced.

The previously established elementary Bessel bounds imply

$$h_g(s) = O(s^4) \quad (0 < s \le 1), \qquad h_g(s) = \alpha_g + O(s^{-1}) \quad (s \ge 1).$$

Thus $\int_0^1 |h_g(s)|\,ds/s$ and $\int_1^\infty |h_g(s) - \alpha_g|\,ds/s$ are finite. Splitting (15) at one proves $\alpha_g \log(K^2) + O(1)$, including $K = 1$. □

This is the scalar determinant in the exact covariant mean-zero decomposition. The earlier globally point-pinned scalar determinant has a different logarithmic coefficient; its compensating vector factor is different as well. Only each complete, exactly normalized determinant combination is invariant under those changes of gauge decomposition. This theorem alone does not compute the vector coefficient or prove positive total drift.

3 Uniform actual-base heat comparison from nonlinear Ward jets

This section concerns every applicable joint infinite structural system with the established homogeneous classical inputs below. Constants are uniform in the old classical depth M. We do not interpolate a finite momentum grid or identify finite winding kernels with periodizations.

Let S_M be the actual homogeneous classical minimum action in current unit coordinates, with a fixed $4 \dim \mathfrak{g}$ input fiber. Use its translation, hypercubic, reality and nonlinear gauge Ward identities, and its uniformly exponentially rooted kernels through field order four. The needed finite spatial moments, local exponential-chart derivatives and their constants are uniform in M. Write $S_W = S_1$ for the normalized Wilson action. The established flat normalization and lower bound are

$$q_M \ge q_W = d_{\mathrm{curl}}^T d_{\mathrm{curl}}, \qquad q_M D = 0, \qquad q_{M,\mathrm{mid}}(p) = |p|^2 I - pp^T + O(|p|^4). \tag{16}$$

These are the actual classical current-unit bounds. A bound for an arbitrary quantum activity with constants depending on a newly chosen auxiliary grouping would not replace them.

The background-Feynman vector base, in the actual left-relative link chart, is

$$\mathcal{B}_M(a) = S''_M(\exp(ia)) + D_a D_a^T.$$

For $\tau(H^2) = 1$ and distinct directions α, ν, define its affine heat coefficient by its convergent fixed-time Taylor kernels:

$$h_M(s) = [t^2]\,\mathrm{tr}_E\big(e^{-s\mathcal{B}_M(a(t))}\big)_{00}, \qquad a_\alpha(t,x) = t x_\nu H. \tag{17}$$

This is an infinite-lattice density, not a varying-flux family on a finite torus.

Theorem 3.1 (Uniform base comparison)**.** *Under these inputs there is C, independent of M, such that*

$$|h_M(s) - h_W(s)| \le C s^{-1} \quad (s \ge 1), \qquad \int_1^{K^2} |h_M(s) - h_W(s)|\,\frac{ds}{s} \le C. \tag{18}$$

The scalar base $D_a^T D_a$ is identical for M and W. Thus the same comparison applies to the vector-minus-two-scalar combination. Explicit periodizations may also be used where their identification has been established.

3.1 Actual relative vertices, including chart curvature

Set $R_M = S_M - S_W$. Its Maxwell coefficient vanishes by (16). The previously proved total nonlinear Ward extraction therefore gives, with uniform analytic moment bounds,

$$R_2 = O(|p|^4), \qquad R_3 = O((\max_i |p_i|)^3), \qquad R_4 = O((\max_i |p_i|)^2). \tag{19}$$

Analytic changes between midpoint and source-site bond phases preserve these orders.

For $\Xi_a(B) = \log(e^{iB}e^{ia})/i$, write

$$J(a) = D_B\Xi_a(0) = I + J_1(a) + J_2(a,a) + O(a^3), \qquad T(a) = D_B^2\Xi_a(0) = T_1(a) + O(a^2).$$

In particular $T(0) = 0$. The exact relative Hessian is

$$L_R(a)[B,C] = R''_M(a)[J(a)B, J(a)C] + R'_M(a)[T(a)[B,C]]. \tag{20}$$

At background degree one it is

$$R_3[a,B,C] + R_2[J_1(a)B, C] + R_2[B, J_1(a)C].$$

Its degree-two coefficient is

$$\begin{aligned}&\tfrac{1}{2}R_4[a,a,B,C] + R_3[a, J_1(a)B, C] + R_3[a, B, J_1(a)C]\\ &\quad + R_2[J_2(a,a)B, C] + R_2[B, J_2(a,a)C] + R_2[J_1(a)B, J_1(a)C] + R_2[a, T_1(a)[B,C]].\end{aligned} \tag{21}$$

Mixed derivatives are obtained by polarization. The last term is the current times chart-curvature term and is retained.

All coefficients of J_1, J_2, T_1 are local at one bond with fixed color norms. They only merge momenta and introduce no inverse momentum. Thus the flat, first-background and second-background differences of the actual Hessians have respective momentum orders $4, 3, 2$. The gauge-fixing term $D_a D_a^T$ is the same for both actions and cancels from these differences.

3.2 Fourier bounds and exact total heat variations

Put $B_M(k) = \mathcal{B}_M(0,k)$ and $\delta(k) = 4\sum_\mu \sin^2(k_\mu/2)$. Then

$$B_W(k) = \delta(k)I, \qquad c\delta(k)I \le B_M(k) \le C\delta(k)I. \tag{22}$$

The upper bound follows from the uniform quadratic zero and bounded second moments. On a fixed small punctured ball, with $\rho = |k|$,

$$\|\partial^r(B_M - B_W)\| \le C\rho^{4-r} \ (r \le 2), \qquad \|\partial B_M\| \le C\rho, \qquad \|\partial^2 B_M\| \le C. \tag{23}$$

Let $V_a^M(k,p)$ be the first vertex for a background $e^{ipx}e_a$, mapping quantum momentum k to $k+p$. Let $W_{ab}^M(k;p,-p)$ be the mixed second vertex, preserving k. For $\Delta = M - W$, (19)–(21) imply at $p = 0$, for $r \le 2$,

$$\begin{aligned} \|\partial_p^r \Delta V_a(k,p)\| + \|\partial_p^r \Delta V_b(k+p,-p)\| &\le C\rho^{3-r}, \\ \|\partial_p^r \Delta W_{ab}(k;p,-p)\| &\le C\rho^{2-r}. \end{aligned} \tag{24}$$

The second first-vertex term includes differentiation of both arguments. For either baseline it suffices to use

$$\|\partial_p^r V\| \le C\rho^{1-r}, \qquad \|\partial_p^r W\| \le C\rho^{-r}. \tag{25}$$

Negative powers here are only loose bounds on analytic functions. In particular $\partial_p^2 W$ is bounded. Away from the small ball all required vertices and derivatives are uniformly bounded.

For compact source variations the second total heat variation is well-defined by trace-class Duhamel insertions. There is no need to assign a value to the extensive vacuum trace. Differentiating $D \operatorname{Tr} e^{-sB}[a] = -s \operatorname{Tr}(e^{-sB}B'[a])$ gives

$$D^2 \operatorname{Tr} e^{-sB}[a,b] = -s \operatorname{Tr}(e^{-sB}B''[a,b]) + s^2 \int_0^1 \operatorname{Tr}(e^{-s(1-v)B}B'[a]e^{-svB}B'[b])\, dv. \tag{26}$$

The integral already includes both orders; there is no further factor two. Only the *total* trace was cycled.

The external-momentum kernel is accordingly

$$\begin{aligned} \mathcal{H}_{ab}^M(s,p) = &-s \int_{\mathrm{BZ}} \operatorname{tr}(e^{-sB_M(k)} W_{ab}^M(k;p,-p)) \frac{dk}{(2\pi)^4} \\ &+ s^2 \int_0^1 \int_{\mathrm{BZ}} \operatorname{tr}\left(e^{-s(1-v)B_M(k+p)} V_a^M(k,p) e^{-svB_M(k)} V_b^M(k+p,-p)\right) \frac{dk}{(2\pi)^4}\, dv. \end{aligned} \tag{27}$$

At fixed s the kernels have the moments permitting two external derivatives under this integral.

3.3 Endpoint-safe heat estimates and the gain

Duhamel differentiation and (22) give, for $E_M(t,k) = e^{-tB_M(k)}$ and $r = 0,1,2$,

$$\|\partial_k^r E_M(t,k)\| \le C_r \rho^{-r}(1+t\rho^2)^r e^{-ct\rho^2}. \tag{28}$$

The sharper first and second derivative factors are respectively $Ct\rho$ and $C(t+t^2\rho^2)$. No spectral frame is differentiated. The exact difference identity

$$E_M(t) - E_W(t) = -\int_0^t E_M(t-u)(B_M - B_W)E_W(u)\, du$$

and (23) give

$$\|\partial_k^r(E_M - E_W)(t,k)\| \le C_r t\rho^{4-r}(1+t\rho^2)^r e^{-ct\rho^2}. \tag{29}$$

These bounds hold for every $t \ge 0$; in particular they contain no singularity at either endpoint $v = 0, 1$ of (27).

Differentiate (27) twice externally and subtract. For the tadpole, the two differences are $(E_M - E_W)\partial_p^2 W_W$ and $E_M\partial_p^2(W_M - W_W)$. Before the outer factor s, they are bounded by $Cs\rho^4 e^{-cs\rho^2}$ and $Ce^{-cs\rho^2}$. Their integrals are $O(s^{-2}) + O(s^{-1})$ after that factor.

Telescope the bubble product into its four factor differences. Replacing a vertex gives combined vertex degree $3 + 1 = 4$. After distributing two derivatives between vertices and heat factors, the full product is bounded by

$$C\rho^2 P(s\rho^2)e^{-cs\rho^2},$$

where P is a fixed polynomial. Replacing a heat factor instead gives, by (29), a time at most s and degree $4 + 1 + 1$; the differentiated bound is

$$Cs\rho^4 P(s\rho^2)e^{-cs\rho^2}.$$

Both bounds are uniform in v, since the two heat times sum to s. The four-dimensional integrals of these expressions are $O(s^{-3})$ and $O(s^{-3})$, respectively. The prefactor s^2 therefore gives $O(s^{-1})$. Away from zero, a fixed spectral gap gives a polynomial in s times e^{-cs}. We have proved

$$|\partial_p^2(\mathcal{H}_{ab}^M - \mathcal{H}_{ab}^W)(s,0)| \le C/s, \qquad s \ge 1. \tag{30}$$

3.4 Density normalization and scope

The diagonal color heat density is gauge invariant, and its linear color jet is zero for a semisimple group. The established density Ward/reanchoring identity at fixed s consequently gives

$$h_M(s) = \tfrac{1}{4}\,\partial_{p_\nu}^2\,\mathcal{H}_{\alpha H,\alpha H}^M(s,0). \tag{31}$$

Indeed the Maxwell extraction is both twice the affine quadratic density and one half of this total Hessian derivative. On the diagonal component the source-site and midpoint phases cancel.

For the actual endpoint-K source $a_\alpha = tx_\nu H/K^2$, translation changes the background only by a constant pure gauge. Every diagonal coefficient in a final block is therefore the same. The K^4 sites cancel the K^{-4} from the two source factors. Its block coefficient is exactly (17); no partial-trace cyclicity or finite-torus flux interpolation enters. Equations (30) and (31) prove Theorem 3.1.

This result supplies the uniform old-depth comparison for the base operator. The soft-penalty, infrared, short-time and coarse-determinant comparisons remain separate inputs to an actual determinant assembly. The proof does not by itself construct a finite-regulator running trajectory, a large-field RG history, a continuum measure or a mass gap.

4 Direct Wilson-to-continuum heat comparison

We now evaluate the fixed reference by its actual vertices and diagonal heat kernels. No general one-loop universality theorem is imported. Let $h_{W,v}(s)$ and $h_{W,g}(s)$ be the affine Cartan coefficients of the Wilson background-Feynman vector base and the full scalar covariant Laplacian on $\mathbb{Z}^4$. Normalize

$$\tau(H^2) = 1, \qquad C_H = \mathrm{Tr}_{\mathrm{ad}}(\mathrm{ad}_H^2), \qquad \alpha_g = \frac{C_H}{48\pi^2}.$$

Theorem 4.1 (Fixed-reference heat coefficient)**.** *For $s \geq 1$,*

$$|h_{W,v}(s) - 10\alpha_g| + |h_{W,g}(s) + \alpha_g/2| \leq Cs^{-1/2}. \tag{32}$$

Together with the actual-base comparison, this gives on every applicable joint infinite structural system

$$h_{M,v}(s) - 2h_g(s) = 11\alpha_g + O(s^{-1/2}), \qquad \int_1^{K^2} (h_{M,v}(s) - 2h_g(s))\frac{ds}{s} = 22\alpha_g \log K + O(1), \tag{33}$$

uniformly in the old classical depth M.

4.1 The exact Wilson density fixes the leading vertices

The normalized unit-lattice action is

$$S_W(U) = \sum_{x,\mu<\nu} \left[1 - \tfrac{1}{2}\tau(P_{x,\mu,\nu} + P_{x,\mu,\nu}^{-1})\right].$$

Each unoriented plaquette is counted once. Its left-relative Hessian and scalar covariant gradient define

$$\mathcal{B}_W(a) = S_W''(e^{ia}) + D_a D_a^T, \qquad L_W(a) = D_a^T D_a, \qquad (D_a\phi)_\mu(x) = \mathrm{Ad}_{e^{ia_\mu(x)}}\,\phi(x + e_\mu) - \phi(x).$$

Transposes use the fixed bilinear counting pairing.

Introduce a physical mesh ϵ, with link angle ϵA. For the physical jet

$$z = (a, b, c, e) = (A_\mu, A_\nu, \nabla_{\epsilon,\mu}A_\nu, \nabla_{\epsilon,\nu}A_\mu)$$

at one plaquette, its holonomy is exactly

$$P_\epsilon(z) = e^{i\epsilon a}e^{i\epsilon b + i\epsilon^2 c}e^{-i\epsilon a - i\epsilon^2 e}e^{-i\epsilon b}. \tag{34}$$

Since $P_0 = I$ and $\partial_\epsilon P_0 = 0$,

$$F_\epsilon = \frac{P_\epsilon - I}{\epsilon^2}, \qquad \mathcal{L}_\epsilon = -\tfrac{1}{2}\tau(F_\epsilon^2 P_\epsilon^{-1})$$

extend jointly holomorphically to $\epsilon = 0$. The identity $P + P^{-1} - 2I = (P - I)^2 P^{-1}$ gives the exact action

$$S_\epsilon(A) = \epsilon^4 \sum_{x,\mu<\nu} \mathcal{L}_\epsilon(J_\epsilon A). \tag{35}$$

Direct differentiation of (34) gives

$$F_0 = i(c - e) - [a, b], \qquad \mathcal{L}_0 = \tfrac{1}{2}\tau\big(c - e + i[a, b]\big)^2. \tag{36}$$

On each fixed bounded complex jet polydisc, $\mathcal{L}_\epsilon - \mathcal{L}_0$, together with every fixed field derivative, is $O(\epsilon)$. Joint holomorphy and Cauchy's formula supply this estimate before any momentum limit is taken.

On plane waves with bounded complex physical momenta, $(e^{i\epsilon p_\mu} - 1)/\epsilon = ip_\mu + O(\epsilon)$ jointly analytically. Equations (35) and (36) therefore identify the continuum action as

$$S_c(A) = \tfrac{1}{2}\int \sum_{\mu<\nu} \tau(F_{\mu\nu}(A)^2)\,dx, \qquad F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + i[A_\mu, A_\nu]. \tag{37}$$

In particular, for the ordinary action vertices,

$$\Gamma_n^W(\epsilon p_1, \ldots, \epsilon p_n) = \epsilon^{4-n}\Gamma_n^c(p_1, \ldots, p_n) + O(\epsilon^{5-n}), \qquad n = 2, 3, 4. \tag{38}$$

This follows by differentiating the exact density in plane-wave amplitudes; it holds jointly with the fixed momentum derivatives used below. In midpoint coordinates the bare quadratic correction is fourth order. Source-site phases may give third-order bare cross entries. The full flat Feynman symbol is exactly $\delta(k)I$ and has the fourth-order correction needed below, in either convention.

The physical left-relative chart is

$$\Xi_{\epsilon,A}(B) = \frac{\log(e^{i\epsilon B}e^{i\epsilon A})}{i\epsilon} = A + B + O(\epsilon).$$

Its first B-derivative is $I + O(\epsilon)$, its second is $O(\epsilon)$. The exact relative Hessian includes both terms

$$S''(A)[J_A B, J_A C] + S'(A)[T_A(B, C)].$$

Thus it has the same continuum limit; the current term is retained. At first background order its local corrections multiply Γ_2, hence have momentum order at least two. At second background order the corrections have order at least one from Γ_3, or two from Γ_2 and the current term. They are within the remainders in (38).

Likewise,

$$\epsilon^{-1}D_{\epsilon A} = \epsilon^{-1}(\mathrm{Ad}_{e^{i\epsilon A_\mu}}\,\mathrm{shift}_{\epsilon e_\mu} - I) \longrightarrow D_{c,A,\mu} = \partial_\mu + i\,\mathrm{ad}_{A_\mu}$$

jointly on these jets. Integration by parts in (37) shows that the limiting vector and scalar operators are

$$(\mathcal{B}_c(A))_{\mu\nu} = -D^2_{c,A}\delta_{\mu\nu} - 2i\,\mathrm{ad}_{F_{\mu\nu}}, \qquad L_c(A) = -D^2_{c,A}. \tag{39}$$

The opposite consistent adjoint sign convention gives identical heat traces. The coefficient of the flat Laplacian is one.

4.2 An integrable comparison of the two-point diagrams

Use the same source-site Fourier convention for lattice and continuum. For either scalar or vector their flat symbols are

$$B_W(k) = \delta(k)I, \qquad B_c(k) = |k|^2 I, \qquad \delta(k) = 4\sum_\mu \sin^2(k_\mu/2).$$

On a fixed small ball with $\rho = |k|$, their difference and its derivatives through order two are $O(\rho^{4-r})$. Define $V_a(k,p)$ and $W_{ab}(k;p,-p)$ as the first and second background vertices, as in the actual-base comparison. The joint analytic expansion above gives

$$\begin{aligned}\|\partial_p^r(V_W - V_c)(k,p)|_{p=0}\| + \|\partial_p^r(V_W - V_c)(k+p,-p)|_{p=0}\| &\le C\rho^{2-r},\\ \|\partial_p^r(W_W - W_c)(k;p,-p)|_{p=0}\| &\le C\rho^{1-r}, \qquad r \le 2.\end{aligned} \tag{40}$$

Here V_c is homogeneous of degree one and W_c is constant. The $r = 2$ bound C/ρ on the second line is deliberately loose: the corresponding lattice derivative is analytic and bounded. For either baseline we can use $\|\partial_p^r V\| \le C\rho^{1-r}$ and $\|\partial_p^r W\| \le C\rho^{-r}$.

The exact total heat Hessian kernel is

$$\begin{aligned}\mathcal{H}_{ab}(s,p) = &-s\int \mathrm{tr}(e^{-sB(k)}W_{ab}(k;p,-p))\,\frac{dk}{(2\pi)^4}\\ &+ s^2\int_0^1\int \mathrm{tr}\left(e^{-s(1-v)B(k+p)}V_a(k,p)e^{-svB(k)}V_b(k+p,-p)\right)\frac{dk}{(2\pi)^4}\,dv.\end{aligned} \tag{41}$$

The lattice integral is over the Brillouin zone, the continuum integral over $\mathbb{R}^4$. This identity is obtained from compact-source Duhamel variations, or defines their absolutely convergent two-point kernel. In

the continuum Gaussian smoothing controls the first-order differential insertions at each $s > 0$. Only total-trace cyclicity is used.

For $r \le 2$, direct differentiation or Duhamel gives

$$\begin{aligned} \|\partial_k^r E(t,k)\| &\le C\rho^{-r}(1+t\rho^2)^r e^{-ct\rho^2}, \\ \|\partial_k^r (E_W - E_c)(t,k)\| &\le Ct\rho^{4-r}(1+t\rho^2)^r e^{-ct\rho^2}. \end{aligned} \tag{42}$$

These estimates hold at $t = 0$ as well; there is no inverse heat time.

After two external derivatives, the tadpole vertex difference is bounded by $C\rho^{-1}e^{-cs\rho^2}$. Its integral with the prefactor s is $O(s^{-1/2})$ in four dimensions. The heat-difference part is bounded by $Cs\rho^4 e^{-cs\rho^2}$ before that prefactor, giving $O(s^{-2})$.

Telescope the bubble product into four differences. Replacing a vertex gives combined degree $2 + 1 = 3$. Distributing two external derivatives among all factors therefore gives

$$C\rho P(s\rho^2)e^{-cs\rho^2}.$$

Its integral times s^2 is $O(s^{-1/2})$. Replacing a heat factor instead gives

$$Cs\rho^4 P(s\rho^2)e^{-cs\rho^2},$$

whose integral times s^2 is $O(s^{-1})$. The estimates are uniform in $v \in [0,1]$: the two heat times sum to s, and (42) has no endpoint singularity. Away from the low-momentum ball, both the lattice and continuum integrals are a polynomial in s times e^{-cs}. The continuum tail uses its polynomial vertices and Gaussian decay. Thus

$$|\partial_p^2(\mathcal{H}_W - \mathcal{H}_c)(s,0)| \le Cs^{-1/2}, \qquad s \ge 1. \tag{43}$$

Equivalently one can rescale $k = \xi/\sqrt{s}$; the displayed bounds provide the needed common integrable majorant.

The lattice gauge-invariant diagonal heat density and its zero linear color jet give

$$h(s) = \tfrac{1}{4}\partial_{p_\nu}^2 \mathcal{H}_{\alpha H,\alpha H}(s,0), \qquad \alpha \ne \nu. \tag{44}$$

Continuum source kernels may contain contact distributions, so we do not assume absolute continuum spatial moments or invoke a continuum reanchoring theorem. The direct transverse diagram calculation below identifies one quarter of the continuum derivative with its explicit magnetic density. This and (43) compare the affine coefficients with error $Cs^{-1/2}$.

4.3 Exact continuum coefficient and sign

Let q range over all eigenvalues of ad_H on the complexified adjoint space, with both charge signs and their multiplicities. For one constant curvature $F_{12} = fH$, the scalar diagonal heat kernel in this charge sector is

$$k_q(s;f) = (4\pi s)^{-2}\frac{qfs}{\sinh(qfs)}. \tag{45}$$

Indeed Fourier transformation in x_2 gives oscillators with eigenvalues $|qf|(2n+1)$; integration over their centers contributes $|qf|/(2\pi)$. Summing the levels and multiplying by the two free directions proves (45) as a pointwise density.

The spin matrix in (39) has eigenvalues $0, 0, 2qf, -2qf$, so the vector density is

$$k_q(s;f)\{2 + 2\cosh(2qfs)\}.$$

Expanding at $f = 0$ for each fixed $s > 0$ gives

$$[f^2]k_q = -\frac{q^2}{96\pi^2}, \qquad [f^2](\text{vector density}) = \frac{10q^2}{48\pi^2}. \tag{46}$$

This uses $x/\sinh x = 1 - x^2/6 + O(x^4)$ and $2 + 2\cosh(2x) = 4 + 4x^2 + O(x^4)$. Complexification has the same trace as the original real color space, so summing charges introduces no extra factor two. Equations (43)–(46) prove (32).

For the required direct identification of the continuum derivative, take an external a_μ with momentum $p = p_\nu e_\nu$, $\mu \neq \nu$. The scalar continuum first vertex is $2qk_\mu$, while its second vertex $2q^2$ has no external derivative. The bubble is

$$2q^2 s(4\pi s)^{-2} \int_0^1 e^{-sv(1-v)p^2}\, dv.$$

Its second p-derivative is $-q^2/(24\pi^2)$, giving the scalar value in (46) after the factor $1/4$. The vector orbital term is four copies of this scalar term. The two spin insertions have trace product $8q^2p^2$, adding $q^2/(4\pi^2)$ to the affine coefficient and yielding the vector value above. Mixed spin/orbital terms have zero Lorentz trace. No spatial-moment assertion about contact distributions is needed.

For the determinant combination $\log\det G - \frac{1}{2}\log\det V$, the proper-time quadratic density is $\frac{1}{2}h_v - h_g$. Extraction with $\beta = 2[t^2]$ therefore gives $h_v - 2h_g$, with this sign and no further multiplier. Together with the actual-base bound, its middle band is $11\alpha_g \log(K^2) + O(1)$, proving (33). For $\tau = \mathrm{Tr}_{\mathrm{fund}}/N$ on $SU(N)$, $C_H = 2N^2$.

This is a positive-time density comparison. It does not assert the existence of separate massless infinite-volume determinants at nonzero magnetic field. Actual soft factors, their joint determinant identities and bounded outer-band corrections must still be assembled in their proved domains; no finite-regulator trajectory or continuum existence is inferred from this reference computation alone.

5 The actual logarithm and a positive structural margin

Return to the exactly normalized determinant correction $\mathcal{D}_{M,K}$ of Notebooks 32–36. All stationary substitutions, coarse constraint factors, and gauge-density factors are those in that construction. The following conclusion combines estimates for that object; it is not a replacement of its determinant by a reference.

Theorem 5.1 (Uniform actual leading coefficient)**.** *Under the classical structural, determinant-identification and localized-vertex inputs stated in this notebook and Notebook 36, there is a constant C_{cmp}, independent of M, K and of the applicable joint infinite structural system, such that*

$$|\beta(\mathcal{D}_{M,K}) - 22\alpha_g \log K| \le C_{\mathrm{cmp}}, \qquad \alpha_g = \frac{C_H}{48\pi^2}. \tag{47}$$

The same constant works for every jointly obtainable structural limit supplying these uniform inputs, including limits with varying old depth. This is not an exact continuous-momentum identity at finite period.

Proof. Write, with the endpoint-K affine normalization,

$$h^{\mathrm{soft}}_{M,K}(s) = [t^2]\{\mathrm{Tr}_{0,E}\, e^{-sV(t)} - 2\,\mathrm{Tr}_{0,S}\, e^{-sT_s(t)}\}.$$

The exact soft factorization, its two bounded coarse corrections, and the two heat-end estimates of Notebook 36 give

$$\beta(\mathcal{D}_{M,K}) = \int_1^{K^2} h^{\mathrm{soft}}_{M,K}(s)\, \frac{ds}{s} + R_{M,K}, \qquad |R_{M,K}| \le C_0. \tag{48}$$

Theorem 1.1 removes the soft penalties on this interval:

$$|h^{\mathrm{soft}}_{M,K}(s) - h^{\mathrm{base}}_{M}(s)| \le C_1 s/K^2, \qquad 1 \le s \le K^2.$$

The scalar base is independent of M. Theorem 3.1 compares the actual vector base with the Wilson base with error at most C_2/s. Theorem 4.1 gives its normalized continuum limit with error at most $C_3 s^{-1/2}$. Consequently

$$|h^{\mathrm{soft}}_{M,K}(s) - 11\alpha_g| \le C\{s^{-1/2} + s/K^2\}, \qquad 1 \le s \le K^2. \tag{49}$$

Both errors are integrable uniformly against logarithmic time:

$$\int_1^{K^2} s^{-1/2}\frac{ds}{s} \le 2, \qquad \int_1^{K^2} \frac{s}{K^2}\frac{ds}{s} \le 1.$$

The constant part integrates to $11\alpha_g \log(K^2) = 22\alpha_g \log K$. Together with (48), this proves (47). At $K = 1$ the intermediate integral is empty and the same bounded-end argument applies.

Every bound used above depends only on the common classical and localized-vertex constants. No uniqueness or favorable selection of a structural limit is used. This proves the stated uniform scope. □

Combining Theorem 2.1 with the exact mean-zero identity $\mathcal{D} = \mathcal{N}_0(\log\det \widehat{G} - \frac{1}{2}\log\det Q_B)$ also gives

$$\beta(\mathcal{N}_0 \log\det \widehat{G}) = 2\alpha_g \log K + O(1), \qquad \beta(-\tfrac{1}{2}\mathcal{N}_0 \log\det Q_B) = 20\alpha_g \log K + O(1).$$

These separated values concern this particular exact decomposition. The complete coefficient (47) retains the original normalization.

For a compact simple group and a nonzero normalized Cartan element, $C_H > 0$. Fixing the elementary quantum ratio $R > 1$, sufficiently large fixed multiples q of the activity-return length therefore give a positive margin for the long ratio $K = R^q$, uniformly in all the structural limits required by finite transfer. The next section states exactly what this supplies for the local quantum recursion. Neither the present coefficient nor that recursion identifies the full Wilson measure or constructs continuum fields.

6 From a uniform positive coefficient to finite local trajectories

This section isolates the exact downstream consequence of a uniform actual determinant comparison. The finite compactness, moment passage, activity induction and endpoint shooting inputs are those of Notebooks 31–32. The coefficient comparison below is an explicit hypothesis of this section; it must be supplied by the actual scalar and vector comparison, with the stated uniformity. No perturbative reference coefficient is substituted for an unidentified actual one.

6.1 Fixed data and the uniform coefficient hypothesis

Fix the elementary quantum ratio $R > 1$, grouping J, field radius $b > 0$, carrier exponent a_0, and the real extracted finite activity system. Let q_a be its amplitude block length, $r_b = K_{\mathrm{amp}} b^2$ its endpoint ball radius, and $C_{\mathrm{int}} b^2$ its intermediate bound. These data are fixed before the longer block length used below. The activity induction holds for arbitrary prescribed couplings in one fixed weak range, provided the retained periods exceed its threshold.

All finite extraction functionals β_j are the actual short-lift functionals normalized by the actual classical reference. For the actual finite local map $F_{j,g}$, write

$$d_j(g,x) = \beta_{j+1}(F_{j,g}(x)), \qquad x_{j+1} = P_{j+1}F_{j,g_j}(x_j), \qquad \gamma_{j+1} = \gamma_j - d_j(g_j, x_j). \tag{50}$$

Real initial classes and the real branches are understood throughout. The initial class may be zero, or any fixed real extracted class in the endpoint ball. The actual coupling relation is $g_j = \gamma_j^{-1/2}$ whenever $\gamma_j > 0$.

Let $D^*_{M,K}$ denote the actual leading determinant correction at preceding homogeneous depth M and endpoint ratio K in a joint infinite structural system. A star labels that structural system, not a preferred limit of one fixed old depth. Its Maxwell functional is the true infinite one, with the same classical normalization used in the finite moment passage. The system includes the shifted depths and endpoint ratios needed by the determinant cocycle. In particular all comparisons below refer to the same joint system.

Uniform actual coefficient comparison UC. Assume that there are $\alpha_g > 0$, $C_{\mathrm{cmp}} \geq 0$ and $K_0 \geq 1$, independent of preceding depth and of the applicable joint structural limit, such that

$$\left|\beta^*(D^*_{M,K}) - 22\alpha_g \log K\right| \leq C_{\mathrm{cmp}} \quad \text{for every } K = R^n \geq K_0. \tag{51}$$

For the one-color convention used in the scalar comparison, $\alpha_g = C_H/(48\pi^2)$, with $C_H > 0$ for a nonzero probe in the simple color factor. Only the uniform lower half of (51) is required for this section. Positivity in one chosen fixed-depth limit is insufficient.

In the present assembly, Theorem 5.1 supplies UC, with $K_0 = 1$, under its stated classical structural and determinant inputs. The following argument retains the hypothesis as a named interface so that its downstream use can be checked independently.

6.2 A positive leading block in every structural limit

Choose a positive multiple q of q_a such that

$$R^q \geq K_0, \qquad C_{\mathrm{cmp}} \leq 11\alpha_g q \log R, \qquad \delta = 11\alpha_g q \log R > 0. \tag{52}$$

The exact leading cocycle and true infinite classical marginal preservation give, in every relevant joint system,

$$B^*_q := \sum_{j=0}^{q-1} \beta^*_{j+1}(D^*_j) = \beta^*_q(D^*_{0,R^q}) \geq 22\alpha_g q \log R - C_{\mathrm{cmp}} \geq \delta. \tag{53}$$

The index zero on the right means the beginning of this block; its preceding homogeneous depth is arbitrary. The cocycle identity is used only in the joint infinite system. No exact additivity of finite short-lift ratios is assumed.

Let $\mathcal{P}$ be the compatible cofinal family of periods. Every fixed power of R, and every required fixed grouping, divides all sufficiently late periods. For a q-step block with final period N, write

$$M_j = MR^j, \qquad Q_j = R^{q-j}N, \qquad b^{M,N}_{0,j} = \beta_{M_jR,Q_{j+1}}(D_{M_j,R;Q_{j+1}}), \qquad B_q(M,N) = \sum_{j=0}^{q-1} b^{M,N}_{0,j}. \tag{54}$$

The available joint compactness and normalized moment passage imply that, along a subsequence of every sequence M_n and $N_n \in \mathcal{P}$ with $N_n \to \infty$,

$$B_q(M_n, N_n) \longrightarrow B^*_q. \tag{55}$$

This passage includes the long-ratio determinant in (53); it does not select unrelated favorable limits. Exponential source-Hessian bounds control the second-moment tails, and the actual finite reference denominators tend uniformly to one.

Lemma 6.1 (Uniform positive finite leading block)**.** *Under* UC*, there is* N_{drift}*, independent of the preceding depth* M*, such that*

$$B_q(M,N) \geq \delta/2 \quad \textit{whenever } N \in \mathcal{P}, \quad N \geq N_{\mathrm{drift}}. \tag{56}$$

Proof. Otherwise choose violating pairs (M_n, N_n) with $N_n \to \infty$. Joint compactness and (55) produce a structural limit with $B_q^* \leq \delta/2$, contradicting (53). The starting stage of a block is absorbed into M. This is a qualitative compactness argument and supplies no numerical period threshold or convergence rate. □

6.3 From leading drift to the actual finite decrement

The finite leakage, zero-input parity and inherited-input comparison give, on the established activity envelope,

$$\begin{aligned} |d_j(g,x_j) - b_{0,j}| \leq C_{\mathrm{int}}\omega(N_{\mathrm{min}}) + C_{\mathrm{par},\beta}\{\tau(T(g)) + g^2T(g)^6\} \\ + C_{\mathrm{in},\beta}\, gT(g)/b. \end{aligned} \tag{57}$$

Here $\omega(N) \to 0$ along the compatible family, and the last two terms tend to zero as $g \downarrow 0$, at the fixed geometric data and field radius. The cutoff $T(g)$ is the actual continuous real threshold. The input factor b^2 has already canceled the b^{-2} Maxwell-extraction bound in the last term.

Choose N_0 above N_{drift}, the reference and finite-induction thresholds, and sufficiently large that

$$C_{\mathrm{int}}\omega(N_0) \leq \frac{\delta}{8q}. \tag{58}$$

Lower the weak-coupling ceiling $g_{\mathrm{w}} > 0$, if necessary, so that for every $0 < g \leq g_{\mathrm{w}}$,

$$C_{\mathrm{par},\beta}\{\tau(T(g)) + g^2T(g)^6\} + C_{\mathrm{in},\beta}gT(g)/b \leq \frac{\delta}{8q}. \tag{59}$$

These choices are independent of the permitted history length. The existing magnitude bound gives a fixed $D_{\mathrm{abs}} > 0$ with $|d_j(g,x_j)| \leq D_{\mathrm{abs}}$ on this envelope. Consequently every q-step block at arbitrary prescribed weak couplings, started in the endpoint ball and with final period at least N_0, obeys

$$B_- \leq \sum_{j=k}^{k+q-1} d_j(g_j,x_j) \leq B_+, \qquad B_- = \delta/4 > 0, \qquad B_+ = qD_{\mathrm{abs}}. \tag{60}$$

Each q_a-step subblock returns to the same endpoint ball. Thus taking q larger repeats the existing activity induction; it does not introduce a new large-endpoint field-dilation estimate. Neither individual leading coefficients nor individual actual decrements need be positive.

6.4 Finite local endpoint shooting

Theorem 6.2 (Conditional positive-drift local trajectories)**.** *Assume* UC *with the stated structural uniformity and the established finite local inputs above. Put* $\Gamma_{\mathrm{w}} = g_{\mathrm{w}}^{-2}$*, and choose a terminal inverse coupling*

$$\Gamma_* \geq \Gamma_{\mathrm{w}} + qD_{\mathrm{abs}}. \tag{61}$$

For every integer $m \geq 1$ *and every compatible finite homogeneous history of* mq *steps with final period* $N_{\mathrm{end}} \geq N_0$*, there is an initial inverse coupling*

$$\Gamma_* + mB_- \leq \gamma_{\mathrm{bare}} \leq \Gamma_* + mB_+ \tag{62}$$

whose actual local extraction trajectory reaches Γ_* *and stays in the weak range throughout. Its initial period is* $R^{mq}N_{\mathrm{end}}$*; all intermediate periods obey the same compatibility requirements. Its activity classes obey the same endpoint and intermediate* $O(b^2)$ *bounds.*

Proof. For the proof alone, evaluate the local map at

$$g_{\mathrm{eval}}(\gamma) = \max\{\gamma, \Gamma_{\mathrm{w}}\}^{-1/2}, \tag{63}$$

while retaining the unclipped scalar update $\gamma_{\mathrm{next}} = \gamma - d_j(g_{\mathrm{eval}}(\gamma), x)$. The evaluated couplings are always weak, so finite induction and (60) apply to every auxiliary history, regardless of the scalar value. The prescribed-coupling estimates apply to its realized sequence; no adaptation changes their constants.

For this fixed finite regulator the joint coupling/class continuity of the normalized local map makes the terminal scalar continuous in its initial value. On the compact interval (62), the absolute increment bound keeps all intermediate scalars bounded. Their evaluated couplings therefore lie in a compact positive interval. No continuity at $g = 0$, volume-uniform continuity modulus, or continuous choice of equivalent activity representatives is needed.

At the left endpoint of (62), the terminal scalar is at most Γ_*, by the lower block bound. At its right endpoint it is at least Γ_*, by the upper block bound. The intermediate value theorem selects a clipped history ending exactly at Γ_*. Summing its future block decrements backwards gives

$$\Gamma_* + (m-k)B_- \le \gamma_{kq} \le \Gamma_* + (m-k)B_+, \qquad 0 \le k \le m. \tag{64}$$

Inside each block,

$$\gamma_{kq+r} \ge \gamma_{kq} - rD_{\mathrm{abs}} \ge \Gamma_* - qD_{\mathrm{abs}} \ge \Gamma_{\mathrm{w}}, \qquad 0 \le r \le q. \tag{65}$$

Clipping is therefore absent from the selected history. Its actual updates are exactly (50) with $g_j = \gamma_j^{-1/2}$. The finite quotient relocalizations replace representatives only by equal normalized totals; their scalar differences are carried by the exact vacuum factors. Thus the conclusion concerns the genuine local density update, not an approximate scalar recurrence. □

The bare coupling in this theorem satisfies

$$(\Gamma_* + mqD_{\mathrm{abs}})^{-1/2} \le g_{\mathrm{bare}} \le (\Gamma_* + m\delta/4)^{-1/2}. \tag{66}$$

It is of order $m^{-1/2}$, equivalently of order $(mq)^{-1/2}$ with q fixed, and reaches the same positive weak endpoint $g_* = \Gamma_*^{-1/2}$. The theorem asserts neither uniqueness of the tuned initial value nor consistency of the choices at different horizons.

6.5 Parameter order and scope

First fix R, J, b and the available q_a-block activity system. The uniform coefficient comparison determines a sufficiently large fixed multiple q by (52). Then choose the period threshold N_0, the weak ceiling g_{w}, and the fixed terminal margin Γ_*. The horizon m is arbitrary after these choices. No quantitative finite-volume rate and no exact finite classical marginal identity enter this order.

The only new coefficient input of this section is (51). If an actual comparison theorem supplies it in every required joint structural system, this section discharges the positive-drift hypothesis of finite local shooting. The comparison is supplied here by Theorem 5.1. The conclusion remains about homogeneous local analytic histories and their justified continuation. It does not identify those histories with the full Wilson measure, control mixed or large-field sectors, construct continuum observable measures, prove a nontrivial continuum Yang–Mills theory, or establish a mass gap.

7 What must still become a four-dimensional quantum theory

The new coefficient removes a local ultraviolet obstruction. It does not turn the analytic local recursion into the full Wilson measure. The next construction must identify that recursion, with its retained windows, gauge insertion, coupling extraction and vacuum convention, inside an exact decomposition of the original finite Wilson integral. Every mixed and large-field history must be included and estimated with compatible boundary conditions. An arbitrary continuation of a small-field germ outside its domain would define a different model.

The next priority is therefore a full-density matching theorem for one fixed scheme. A useful statement would start from the actual positive finite Wilson measure and express its pushforward as the proved local sector plus explicitly controlled complementary sectors, at the tuned couplings just constructed. It must hold uniformly as the ultraviolet horizon grows. This statement is a target, not a consequence of the one-loop calculation.

External observable sources then require their own uniform estimates. Holomorphy in a connection background does not establish holomorphy in a source coupled to a renormalized observable. Source counterterms, normalization denominators, reflection-cell domains, and estimates of all needed orders must be supplied for the same full measure. Those estimates must produce a common continuum hierarchy with compatible products, rather than separate limits of unrelated observables.

Reflection positivity must pass from that same positive Wilson sequence. Full Euclidean covariance, the required distributional growth, a suitable vacuum sector, and reconstruction must be verified. Finally, a nonzero scalar correlation alone does not identify Yang–Mills: the limit needs an adequate mixed family of gauge-invariant curvature observables and their short-distance normalization and transformation laws. An interacting lower bound and the relevant identification remain separate tasks. The mass-gap problem requires an additional physical infrared argument.

The companion file `work/final_push_existence_dependency_audit.txt` records the detailed dependency map and source boundaries. It explicitly distinguishes established working modules, conditional reductions, and the open model-specific inputs. Earlier notebooks remain chronological records; their older next-step paragraphs must be read together with the results of the present notebook.

References

[1] *Toward a four-dimensional Yang–Mills construction*, research Notebooks 15–36 in this workspace. In particular, Notebook 32 supplies finite moment transfer and local shooting; Notebook 33 fixes the continuum normalization and affine density identification; Notebook 34 computes the exact unpinned scalar heat coefficient; Notebooks 35–36 supply the actual mean-zero and soft determinant identities and localized estimates.

[2] D. V. Vassilevich, *Heat kernel expansion: user's manual*, Physics Reports **388** (2003), 279–360, arXiv:hep-th/0306138. The reference coefficient is calculated explicitly here; the source is not a proof of the actual lattice comparison.

[3] D. H. Adams and W. Lee, *Structure of logarithmically divergent one-loop lattice Feynman integrals*, Phys. Rev. D **77** (2008), 045010, arXiv:0709.0781. Its fixed-integrand universality theorem is useful background; uniformity in our varying effective actions is proved directly above.

[4] A. Jaffe and E. Witten, *Quantum Yang–Mills Theory*, Clay Mathematics Institute problem description.

[5] E. Sussman and collaborators, Yang–Mills-LLM-experiment, deposited accounts pinned at commit `e392ef45013a581664bab76b90867170b30fe10e`.